

FAMILY NAME:
OTHER NAME(S):
STUDENT NUMBER:
SIGNATURE:

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF MATHEMATICS AND STATISTICS

Final Exam, T1, 2021

MATH5835
Advanced Stochastic Processes

- (1) TIME ALLOWED – 3 hours (Including time for preparing and submitting your work)
- (2) TOTAL NUMBER OF QUESTIONS – 5
- (3) ANSWER ALL QUESTIONS
- (4) THE QUESTIONS ARE **NOT** OF EQUAL VALUE
- (5) TOTAL NUMBER OF MARKS – 100
- (6) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE

This is an open-book exam, so you are allowed to consult the lecture notes, textbooks, or any other material posted on Moodle. Please handwrite your answers to the questions and then submit a **single PDF** file containing your answers. **Make sure you submit all your answers, and the answers are clearly legible.**

The first page of the PDF file must include a photograph of your **student ID card** with the **signed**, handwritten statement:

“I declare that this submission is entirely my own original work.”

1. [20 marks] Miss Artemis owns $K = 10$ umbrellas, which are either at home or at her office in the city. She goes to work every morning and goes home every evening. If (and only if) it rains when she leaves home or work, she takes an umbrella with her (if one is available). Assume that the probability of raining when she leaves home in the morning or when she leaves work in the evening is always $p = 1/4$, independent of previous weather conditions. Let X_n be the number of umbrellas at home at the end of day n . Assume $X_0 = K = 10$.
- [2 marks] What is the state space of the process $\{X_n, n = 0, 1, 2, \dots\}$.
 - [5 marks] Briefly explain why $\{X_n\}$ is a Markov process, and find its transition probability matrix P .
 - [6 marks] Find the invariant distribution π of $\{X_n\}$.
 - [3 marks] Is the chain reversible in equilibrium?
 - [4 marks] What is the expected number of days to wait until all the 10 umbrellas are at her home again?
2. [20 marks] Let $X_1, X_2, \dots$ be independent and let each X_n have Poisson distribution with parameter λ .

- a) [5 marks] Show that the process

$$M_n = X_1 + \dots + X_n - n\lambda, \quad n \geq 1,$$

is a martingale with respect to the filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, $n = 1, 2, \dots$.

- b) [5 marks] Let

$$\tau = \min \{n \geq 1 : X_1 + \dots + X_n \geq 2\}.$$

Show that τ is a stopping time.

- c) [5 marks] Show

$$\mathbb{P}(\tau > n) = e^{-n\lambda}(1 + n\lambda), \quad n = 0, 1, \dots$$

- d) [5 marks] Show

$$\mathbb{E}S_\tau = \frac{\lambda - \lambda e^{-\lambda} + \lambda^2 e^{-\lambda}}{(1 - e^{-\lambda})^2},$$

where $S_n = X_1 + \dots + X_n$.

3. [10 marks] Let $(M_n, \mathcal{F}_n)$ be a martingale with $M_0 = 0$ and assume that

$$\sum_{n=1}^{\infty} \mathbb{E} (M_n - M_{n-1})^2 < \infty.$$

- a) [5 marks] Show that there exists a random variable M_{∞} such that

$$\mathbb{P}(M_n \rightarrow M_{\infty}) = 1, \quad \mathbb{E} M_{\infty}^2 < \infty.$$

- b) [5 marks] Let $X_1, X_2, \dots$ be independent, and $\mathbb{E} X_n^2 < \infty$ for all $n \geq 1$. Assume also that

$$\sum_{n=1}^{\infty} \mathbb{E} |X_n| < \infty, \quad \sum_{n=1}^{\infty} \mathbb{E} X_n^2 < \infty.$$

Show that the series

$$\sum_{n=1}^{\infty} X_n$$

converges with probability one.

4. [20 marks] Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ which is Poisson distributed with mean $\lambda > 0$. Let $\mathbb{Q}$ be the set function defined by $\mathbb{Q}(A) = \mathbb{E}[e^{X \log \lambda - \lambda^2 + \lambda}; A]$, for $A \in \mathcal{F}$.

- a) [5 marks] Recall that the moment generating function of a random variable ξ is defined by $M_{\xi}(t) = \mathbb{E}[e^{t\xi}]$, for t in a neighbourhood of 0 where the expectation is finite. Show that the moment generating function of X is given by

$$M_X(t) = e^{\lambda(e^t - 1)}, \quad t \in \mathbb{R}.$$

- b) [5 marks] Show that the set function $\mathbb{Q}$ is a probability measure on $(\Omega, \mathcal{F})$.
 c) [5 marks] Find the distribution of X under the probability measure $\mathbb{Q}$.
 d) [5 marks] Find a probability measure $\mathbb{M}$ under which the distribution of X is Poisson with mean 1.

5. [10 marks] Let B_t , $t \geq 0$ be a standard Brownian motion, N_t , $t \geq 0$ be a Poisson process with rate $\lambda > 0$. Assume $\{B_t; t \geq 0\}$ and $\{N_t; t \geq 0\}$ are independent. Define $X_t = \sigma B_t + N_t$, $t \geq 0$, where $\sigma \in \mathbb{R}$.

- a) [5 marks] Show $M_t = \exp\{\theta X_t - \frac{1}{2}\theta^2\sigma^2 t + \lambda(1 - e^{\theta})t\}$, $t \geq 0$ is a martingale with respect to the filtration $\mathcal{F}_t = \sigma\{B_s, N_s; s \leq t\}$, $t \geq 0$.
 b) [5 marks] Solve the stochastic differential equation

$$dr(t) = \{\theta(t) - \alpha(t)r(t)\} dt + \sigma(t)dB(t),$$

with initial condition $r(0) = r_0$. [Hint: Try to find a deterministic function $A(t)$ such that $d\{A(t)r(t)\}$ is free of $r(t)$.]

END OF EXAM